

## Verification Phase:

The above queries confirm that all tables were successfully populated and that referential integrity between related tables is correctly enforced.

Verification that each table contains 6 records:

```
MariaDB [ak26db]> SELECT COUNT(*) FROM Organization;
```

| COUNT(*) |
|----------|
| 6        |

```
1 row in set (0.000 sec)
```

```
MariaDB [ak26db]>
```

```
MariaDB [ak26db]>
```

```
MariaDB [ak26db]> SELECT COUNT(*) FROM Program;
```

| COUNT(*) |
|----------|
| 6        |

```
1 row in set (0.000 sec)
```

```
MariaDB [ak26db]>
```

```
MariaDB [ak26db]> SELECT COUNT(*) FROM ApplicantUser;
```

| COUNT(*) |
|----------|
| 6        |

```
1 row in set (0.000 sec)
```

```
MariaDB [ak26db]>
```

```
MariaDB [ak26db]>
```

```
MariaDB [ak26db]> SELECT COUNT(*) FROM Application;
```

| COUNT(*) |
|----------|
| 6        |

```
1 row in set (0.000 sec)
```

```
MariaDB [ak26db]>
```

### Full chain (Application → User → Program → Organization) Verification

```
MariaDB [ak26db]> SELECT
->   a.application_id,
->   u.first_name,
->   u.last_name,
->   p.program_name,
->   a.status
-> FROM Application a
-> JOIN ApplicantUser u ON a.user_id = u.user_id
-> JOIN Program p ON a.program_id = p.program_id;
```

| application_id | first_name | last_name | program_name                    | status       |
|----------------|------------|-----------|---------------------------------|--------------|
| 1              | John       | Smith     | Prescription Assistance Program | Pending      |
| 2              | Emily      | Johnson   | Child Health Coverage Plan      | Approved     |
| 3              | Michael    | Brown     | Senior Medical Aid              | Under Review |
| 4              | Sarah      | Davis     | Emergency Treatment Grant       | Rejected     |
| 5              | David      | Wilson    | Maternal Care Assistance        | Pending      |
| 6              | Olivia     | Martinez  | Chronic Illness Support Program | Approved     |

6 rows in set (0.001 sec)

```
MariaDB [ak26db]> |
```

### Applications per program Verification:

```
MariaDB [ak26db]> SELECT
->   p.program_name,
->   COUNT(a.application_id) AS total_applications
-> FROM Program p
-> LEFT JOIN Application a ON p.program_id = a.program_id
-> GROUP BY p.program_name;
```

| program_name                    | total_applications |
|---------------------------------|--------------------|
| Child Health Coverage Plan      | 1                  |
| Chronic Illness Support Program | 1                  |
| Emergency Treatment Grant       | 1                  |
| Maternal Care Assistance        | 1                  |
| Prescription Assistance Program | 1                  |
| Senior Medical Aid              | 1                  |

6 rows in set (0.008 sec)

```
MariaDB [ak26db]> |
```

Application per User Verification:

```
MariaDB [ak26db]> SELECT
->     u.first_name,
->     u.last_name,
->     COUNT(a.application_id) AS total_applications
-> FROM ApplicantUser u
-> LEFT JOIN Application a ON u.user_id = a.user_id
-> GROUP BY u.user_id;
```

| first_name | last_name | total_applications |
|------------|-----------|--------------------|
| John       | Smith     | 1                  |
| Emily      | Johnson   | 1                  |
| Michael    | Brown     | 1                  |
| Sarah      | Davis     | 1                  |
| David      | Wilson    | 1                  |
| Olivia     | Martinez  | 1                  |

```
6 rows in set (0.001 sec)

MariaDB [ak26db]>
```

Edge Case Verification: Programs without applications:

```
MariaDB [ak26db]> SELECT
->     p.program_name
-> FROM Program p
-> LEFT JOIN Application a ON p.program_id = a.program_id
-> WHERE a.application_id IS NULL;
```

Empty set (0.000 sec)

```
MariaDB [ak26db]>
MariaDB [ak26db]>
MariaDB [ak26db]> |
```